



DhanDrishti धन-दृष्टि

*Cash-flow foresight & early-warning intelligence
for rural micro enterprises*

AI-Driven Cash Flow Prediction & Risk Flagging System for Rural Micro Enterprises

NABARD HACKATHON · GLOBAL FINTECH FEST 2026 · ROUND 1

Working prototype included — every number in this deck is measured output.

THE PROBLEM

Bharat's micro enterprises are credit-invisible

14%

of India's 6.3 crore MSMEs
have formal credit access

Deloitte, 2025

₹25L Cr+

MSME financing gap —
worst in rural districts

≈ rural 32% vs urban 20%

Too late

lenders learn of stress only
after an EMI is missed

no forward-looking tooling

Kirana stores, dairy units, tailors, agri-input dealers, food processors, handloom weavers — cash-heavy, seasonal, no audited books, thin bureau files. **Field officers cover hundreds of accounts with no early-warning system: by the time stress is visible, the intervention window has closed.**

The signal now exists — nobody has stitched it together



UPI

≈ 21.7 billion transactions a month — rural merchants included

→ *digital revenue rhythm, counterparty mix*



Account Aggregator

₹1.67 lakh crore disbursed on AA rails in FY25, with user consent

→ *bank-statement cash flows, on tap*



Open public data

Agmarknet mandi prices · eNAM · IMD rainfall · crop calendars

→ *the context that drives rural cash cycles*

The gap: these signals have never been fused into enterprise-level cash-flow foresight for rural Bharat. That is exactly what DhanDrishti does.

Four outputs, one loop: foresee → flag → explain → act



Probabilistic forecast

12-week net cash-flow bands (P10/P50/P90), conformally calibrated — honest uncertainty, not a point guess.



Risk score + flags

Calibrated 0–100 score with discrete early warnings: liquidity crunch ahead, EMI stress, buyer concentration, seasonal dip...



Bilingual reason codes

Every alert explained in English + Hindi from SHAP attributions: “Revenue is down ~58% vs the same season last year.”



Recommended action

One concrete intervention per flag: pre-approve working capital, restructure EMI post-harvest, diversify buyers.

Surfaces: offline-capable dashboard (EN/हिंदी) · scoring API for bank LOS/LMS · SMS/WhatsApp nudges (roadmap)

The command deck — running, offline, bilingual



LIVE ALERT TICKER

Every HIGH/WATCH enterprise scrolls past with its top flag — click any alert to drill straight in.

RISK-RANKED WATCHLIST

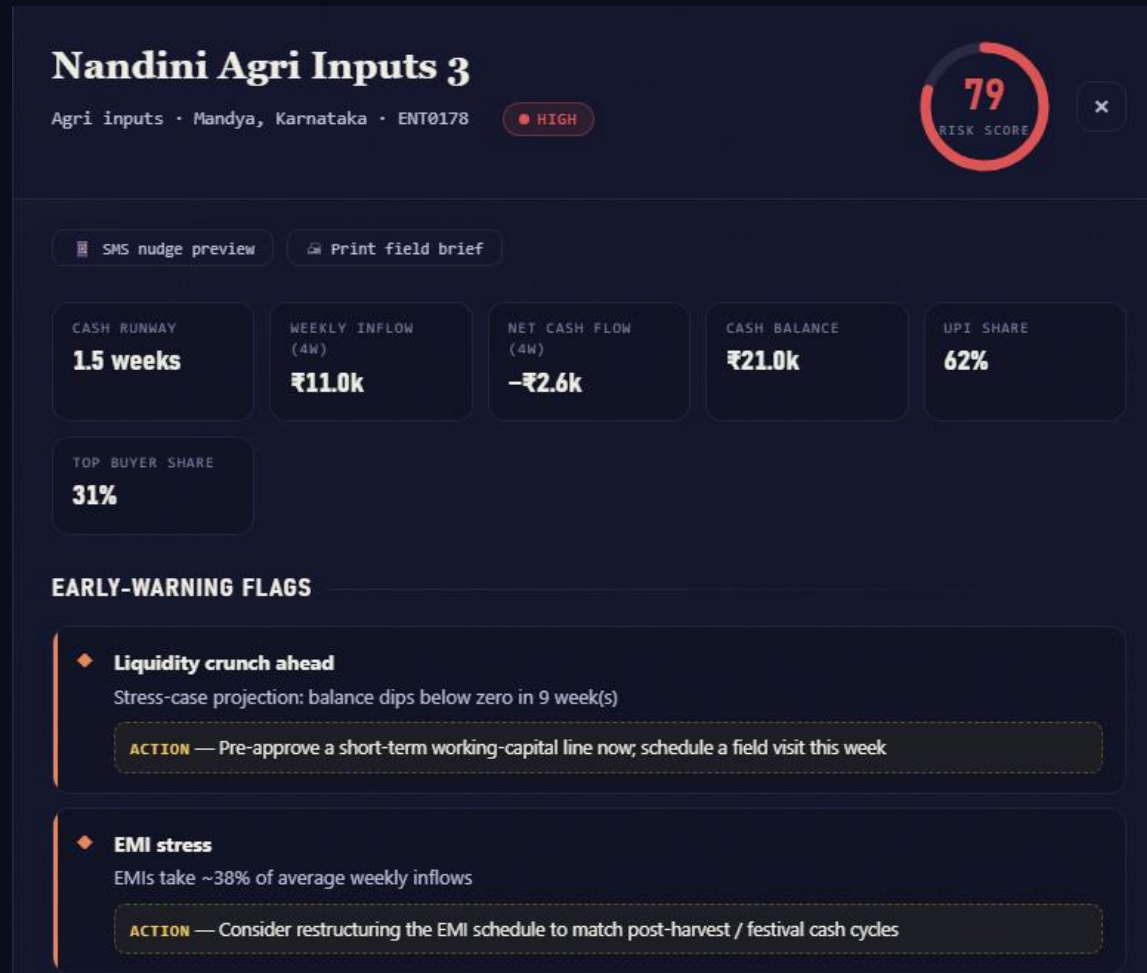
200 enterprises, calibrated 0–100 scores, sparklines and signal chips; search, filter, sort in place.

BILINGUAL & OFFLINE

EN/हिंदी toggle on every reason and action; zero external assets — the demo runs with the internet off.

Deep-link any enterprise for the jury: localhost:8765/#ENT0178

Nine weeks of warning before the balance hits zero



Risk 79/100 — flagged today. The stress path crosses ₹0 in week 9. The field officer sees two flags, four plain-language reasons, and one concrete action: pre-approve a working-capital line now — in Hindi if she prefers. A missed EMI would have surfaced this ~2 months later.



Boring where it should be, clever where it counts



Feature store

28 weekly signals per enterprise: cash-flow trend & volatility, runway, UPI share drift, buyer concentration, EMI burden, mandi & rainfall context, festival calendar.



Quantile forecaster

Gradient-boosted P10/P50/P90, direct multi-horizon (12 wk), conformalized on a temporal holdout — the 80% band actually covers 80%.



Early-warning classifier

Predicts distress within 8 weeks. Domain monotonicity constraints (risk ↑ as runway ↓, EMI ↑, volatility ↑) + Platt calibration = smooth, defensible probabilities.



Explainability layer

TreeSHAP → curated bilingual reason codes. Transparent rules overlay generates flags; a noisy-OR blend guarantees the score and flags never contradict.

CPU-only (LightGBM) · retrains weekly in minutes · runs on-prem at an RRB data centre

Prototype results on a 200-enterprise simulated panel

-25%

forecast MAE vs
seasonal-naive baseline

80%

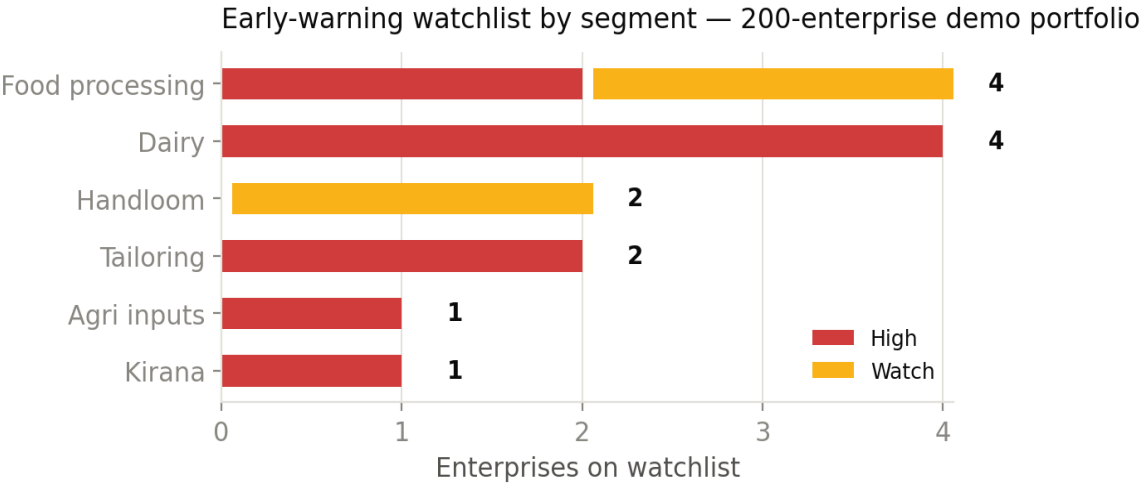
band coverage after
conformal calibration

0.83

early-warning AUC
(13× precision lift)

3 wks

median warning lead
before distress



Every top-10 alert on the demo panel corresponds to a genuine engineered distress ramp. Data is synthetic-but-realistic: harvest & festival seasonality, monsoon shocks, UPI adoption drift, household draws, and ground-truth distress episodes — so accuracy is measurable, honestly.

Consent-first and DPI-native by design



Account Aggregator

bank/UPI statements — inflow rhythm, balances, counterparties (consented pull)



UPI / QR payments

daily revenue proxy, customer count, ticket size



SHG / JLG ledgers · Udyam

group repayment history, enterprise vintage & segment



Agmarknet · eNAM · IMD

mandi prices, rainfall, crop calendar — the rural context layer



No Aadhaar numbers · no location tracking · no scraping. Pseudonymous features, logged consent artefacts, DPDP-aligned. Cold start via segment seasonality priors.

If a BC sakhi can't use it, it doesn't count



Works where the network doesn't

Offline-capable PWA, zero external assets, SMS fallback on the roadmap. Demo runs with the internet switched off.



Speaks the field officer's language

Every reason and action in English + Hindi (extensible to Marathi, Telugu, Bangla...). Not a score — a sentence someone can act on.



Runs on what banks already have

CPU-only models, one server per district pilot, on-prem deployable at an RRB — no cloud dependency, no GPUs.

SAMPLE ALERT — AS THE FIELD OFFICER SEES IT

- ▲ Liquidity crunch ahead — stress-case balance dips below zero in 9 weeks. → Pre-approve a short-term working-capital line; schedule a field visit this week.
- ▲ नकदी संकट की आशंका — तनाव-परिदृश्य में 9 सप्ताह में शेष राशि शून्य से नीचे जा सकती है। → अभी अल्पकालिक कार्यशील-पूंजी सीमा स्वीकृत करें; इसी सप्ताह क्षेत्र भ्रमण तय करें।

One rail, three winners



Lenders (RRBs, DCCBs, MFIs)

- Weeks of early warning → intervene before slippage, cut micro-book NPAs
- Explainable, auditable basis for cash-flow-based lending
- Portfolio heatmaps by district & segment



Rural entrepreneurs

- A bankable, consent-based cash-flow record — the on-ramp from moneylenders to formal credit
- Vernacular nudges to plan for lean seasons
- Credit priced on evidence, not absence of it



NABARD & policy

- District-level early-warning view across the rural portfolio
- Evidence base for scheme design & timing
- A reusable public-good scoring rail for the SHG/RRB ecosystem

From working prototype to district pilot

NOW · DONE

JUL–AUG 2026

POST-GFF

THEN

1

Working prototype

Full pipeline: data → models → bilingual dashboard. Measured: –25% MAE, AUC 0.83, 3-wk lead.

2

Round 2 build

AA-sandbox integration demo, SMS nudge prototype, hardening, demo video + deployment guide.

3

District pilot

One RRB/DCCB partner, 500–1,000 consented enterprises. Measure lead time, intervention uptake, repayment outcomes vs control.

4

Scale via NABARD

State roll-out through the RRB/SHG ecosystem, OCEN-integrated scoring API, more languages.



Seeing cash-flow stress before it strikes.

DhanDrishti · cash-flow foresight & early warning for rural micro enterprises

Team «name» — «members & roles» · «email» · «phone»

Prototype: runs offline on a laptop · synthetic data only · no real enterprise or personal data